

Deep Operator Learning of Maximum-Principle-Preserving Dissipative Semigroups for Time-Dependent PDEs

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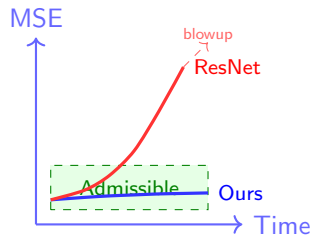
Outline

- 1 Motivation
- 2 Semigroup Formulation
- 3 Latent Architecture
- 4 Theoretical Results
- 5 Numerical Experiments
- 6 Conclusion

The Problem: Learned Operators Break Under Repetition

Standard operator learning (FNO, DeepONet, ResNet):

- Learn a one-step map $\hat{u}^{n+1} = \mathcal{N}_\theta(u^n)$
- Minimize single-step MSE
- **No guarantee** that repeated composition stays physical



Consequence:

- Boundary violations accumulate
- Energy grows instead of decaying
- **Numerical blowup** at long times

Question: Can we design architectures that respect physics *by construction*?

Two Structural Laws of Dissipative PDEs

Running example: Fisher–KPP

$$\partial_t u = \nu \Delta u + ru(1-u) \quad \text{with} \quad u \in [0, 1], \quad \mathcal{E}(u) = \int_{\Omega} \left(\frac{\nu}{2} |\nabla u|^2 - r \left(\frac{u^2}{2} - \frac{u^3}{3} \right) \right) dx \searrow$$

Maximum Principle / Invariant Region

$$u_0 \in \mathcal{K} \implies \mathcal{S}_t u_0 \in \mathcal{K} \quad \forall t \geq 0$$

- Fisher–KPP: $u \in [0, 1]$
- Allen–Cahn: $u \in [-1, 1]$
- Physical concentrations, temperatures, etc.

Energy Dissipation

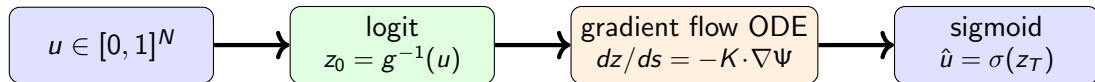
$$\mathcal{E}(\mathcal{S}_t u_0) \leq \mathcal{E}(u_0) \quad \forall t \geq 0$$

- Free energy decreases along trajectories
- Guarantees long-time stability
- Prevents spurious oscillations

Key insight: These hold for the *PDE semigroup* $(\mathcal{S}_t)_{t \geq 0}$. Can we enforce them in a *learned* operator?

Key Insight: Enforce Structure by Design

Idea: Embed the three laws directly into the **network architecture**, so they hold *for every parameter θ* —no training penalty needed.



$$g = \sigma$$

Bounded output
 $\Rightarrow$ admissibility

$$K \succ 0 + \text{gradient flow}$$

Energy decreases
 $\Rightarrow$ dissipation

ODE flow

Unique trajectories
 $\Rightarrow$ semigroup

Why Standard Networks Fail at Long Times

ResNet / FNO trained for one step:

- Minimize $\|\mathcal{N}_\theta(u^n) - u^{n+1}\|^2$
- No structural constraint
- Single-step error is small

But repeated composition:

- $\varepsilon_{\text{sg}} > 0 \Rightarrow$ errors **accumulate**
- Boundary violations $\rightarrow$ unphysical states
- Energy grows $\rightarrow$ spurious oscillations
- Eventually: **numerical blowup**

Fisher-KPP at $T = 20$

Method	BV	Energy
FNO	15.4%	grows
ResNet	blowup	blowup
Ours	0.00	100%

Root cause

$\varepsilon_{\text{sg}} > 0$ in standard networks:

$$\Phi(\Phi(u, \tau_1), \tau_2) \neq \Phi(u, \tau_1 + \tau_2)$$

Structure-Preserving Deep Semigroup Learner

Definition (Target specification)

A learned evolution family $\Phi_\theta(\cdot, \tau) : \mathcal{K} \rightarrow \mathcal{K}$ is **structure-preserving** if:

$$(S1) \quad \Phi_\theta(u, 0) = u \quad \text{(identity)} \quad (1)$$

$$(S2) \quad \Phi_\theta(u, \tau) \in \mathcal{K} \quad \text{(admissibility)} \quad (2)$$

$$(S3) \quad \Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) = \Phi_\theta(u, \tau_1 + \tau_2) \quad \text{(semigroup)} \quad (3)$$

$$(S4) \quad \mathcal{E}(\Phi_\theta(u, \tau)) \leq \mathcal{E}(u) \quad \text{(dissipation)} \quad (4)$$

- (S1)–(S3) define an *evolution family* for repeated composition
- (S4) ensures energy stability
- These are **architecture-level** guarantees: hold for every θ

Error Propagation Under Repeated Composition

Theorem (Error decomposition)

$$\|u^n - \mathcal{S}_{n\Delta t} u_0\| \leq \underbrace{\varepsilon_{\text{app}}}_{\text{approximation}} + n \cdot \underbrace{\varepsilon_{\text{sg}}}_{\text{semigroup defect}}$$

where

- ε_{app} : *trajectory-level fit from u_0*
- $\varepsilon_{\text{sg}} = \sup_u \|\Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) - \Phi_\theta(u, \tau_1 + \tau_2)\|$

Why this matters

If $\varepsilon_{\text{sg}} > 0$ (generic neural network), errors **accumulate linearly** in n . If $\varepsilon_{\text{sg}} = 0$ (exact semigroup), only approximation error remains.

Network Structure: Latent Dissipative Flow

Step 1. Bounded latent map. Let $\sigma(\xi) = (1 + e^{-\xi})^{-1}$. Define $g : \mathbb{R}^N \rightarrow (m, M)^N$ componentwise by

$$g(z)_i = m + (M - m) \sigma(z_i), \quad g^{-1}(u)_i = \log\left(\frac{u_i - m}{M - u_i}\right).$$

- $\sigma : \mathbb{R} \rightarrow (0, 1)$ is C^∞ , strictly increasing
- g maps *all* of $\mathbb{R}^N$ into $(m, M)^N \Rightarrow$ **admissibility by construction**

Step 2. Latent energy (reaction–diffusion ansatz).

$$\Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2$$

- $V_\theta \in C^1(\mathbb{R}; \mathbb{R})$: scalar potential (MLP + $\frac{\beta_V}{2} \xi^2$, $\rho = \text{softplus}$)
- $a_{ij}^{(\theta)} = m_{ij} \text{softplus}(e_i^\top e_j) \geq 0$: learned interaction (m_{ij} : local mask, $e_i \in \mathbb{R}^{r_A}$: embeddings)

Network Structure: Latent Dissipative Flow (cont.)

Step 3. Diagonal PSD mobility.

$$K_{\theta}(z) = \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z)), \quad k_{\theta,i}(z) = \text{softplus}(\text{StencilMLP}(z_{\mathcal{N}(i)})) > 0$$

- StencilMLP: local input $z_{\mathcal{N}(i)}$ (periodic padding), MLP + softplus
- $K_{\theta} \succ 0$ pointwise $\Rightarrow$ **dissipation by construction**

Step 4. Latent gradient-flow ODE.

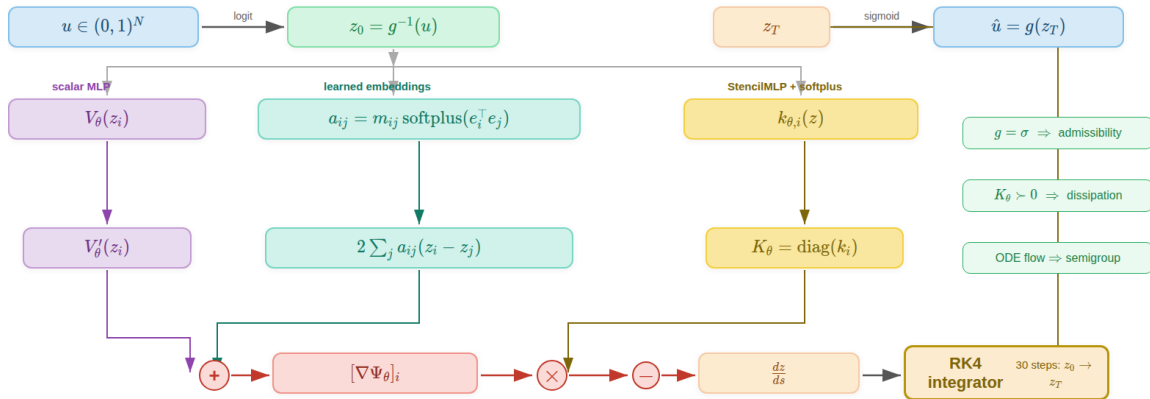
$$\partial_s z_i = -k_{\theta,i}(z) \left(V'_{\theta}(z_i) + 2 \sum_j a_{ij}^{(\theta)}(z_i - z_j) \right), \quad z(0) = g^{-1}(u)$$

- $dz/ds = -K_{\theta}(z) \nabla \Psi_{\theta}(z)$: gradient flow in latent space
- Integrated by RK4 with $M = 30$ substeps per τ

Step 5. One-step operator.

$$\Phi_{\theta}(u, \tau) := g(\varphi_{\theta}^{\tau}(g^{-1}(u))), \quad \varphi_{\theta}^{\tau} = \text{ODE flow for time } \tau$$

Network Architecture



Three Guarantees by Construction

Admissibility

$$u = \sigma(z) \in (0, 1)^N$$



Holds for *all* $z \in \mathbb{R}^N$

Dissipation

$$\frac{d\Psi}{ds} = -K \|\nabla \Psi\|^2 \leq 0$$



$K \succ 0$ by design

Semigroup

$$\Phi_\theta \circ \Phi_\theta = \Phi_\theta$$



ODE flow uniqueness

All three hold for every parameter θ , independent of training.

Neural Parametrization

Scalar potential V_θ

$$V_\theta(z_i) = \text{MLP}(z_i) + \frac{\beta}{2} z_i^2$$

- softplus activation (C^∞)
- $\beta \geq 0$: quadratic growth for coercivity

Mobility K_θ

$$k_{\theta,i}(z) = \text{StencilMLP}(z_{\mathcal{N}(i)}) + \epsilon$$

- Local stencil input (periodic padding)
- softplus $+\epsilon > 0$: positive diagonal

Interaction coefficients

$$a_{ij} = m_{ij} \text{softplus}(e_i^\top e_j) \geq 0$$

- m_{ij} : local interaction mask ($|i-j| \leq r$)
- $e_i \in \mathbb{R}^8$: learned embeddings

Training Loss

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{step}} = \|\Phi_{\theta}(u, \tau) - u^{\text{ref}}\|^2$$

What we optimize

- One-step prediction: match PDE snapshot at $t + \tau$
- Standard MSE loss
- No auxiliary regularization terms

Why no other losses?

- **Semigroup:** enforced by ODE flow uniqueness; RK4 truncation $\sim 10^{-14}$ is negligible
- **Dissipation:** enforced by $K_{\theta} \succ 0$ + gradient flow ($d\Psi/ds \leq 0$)
- **Both verified as post-training diagnostics**

Transfer Theorem

Theorem (Continuous $\rightarrow$ discrete $\rightarrow$ latent)

Under Assumptions 1–3, for initial data $u_0 \in \mathcal{K}$:

$$\left\| \Phi_{\theta}(u_0^N, n\Delta t) - \mathcal{S}_{n\Delta t} u_0 \right\| \leq C(N) + n\Delta t \varepsilon_{\text{match}}$$

where

- $C(N)$: spatial discretization rate (Proposition 8)
 - $\varepsilon_{\text{match}}$: generator matching error
-
- Decomposes the chain: PDE $\rightarrow$ semidiscrete $\rightarrow$ latent architecture
 - $C(N) \rightarrow 0$ as $N \rightarrow \infty$ (consistent spatial discretization)
 - $\varepsilon_{\text{match}} \rightarrow 0$: universal approximation of the generator

Expressivity: Universal Approximation

Proposition (Gradient-type generators)

For any smooth semidiscrete generator $\mathcal{F}_N$ of reaction–diffusion type, the latent architecture can approximate $\mathcal{F}_N$ to arbitrary accuracy on compact subsets of $(m, M)^N$.

Proof structure (4 steps):

- 1 MLP approximation of scalar potential V
- 2 Embedding approximation of interaction coefficients a_{ij}
- 3 Stencil MLP approximation of mobility K
- 4 Product composition with Lipschitz stability

Scope

- **Applies to:** Fisher–KPP, Allen–Cahn (gradient-type generators)
- **Does not formally apply to:** Burgers (skew-symmetric convective term)
- Good empirical results for Burgers suggest expressivity beyond gradient class

Summary: What is Proved vs. What is Open

Property	Level	Status
Admissibility $u \in (0, 1)^N$	Architecture	Exact
Energy dissipation	Architecture	Exact
Semigroup composition	Architecture	Exact
Spatial convergence $C(N) \rightarrow 0$	Semidiscrete	Proved
Generator approximation	Latent	Proved
Global flow existence ($\beta_V = 0$)	Architecture	Open
PDE-level transfer	Full chain	Open

Architecture-level guarantees are unconditional. Transfer to PDE level is formulated as open problems.

Experimental Setup

Three PDE Benchmarks

- 1 **Fisher–KPP**: $\partial_t u = \nu \Delta u + ru(1-u)$,
 $u \in [0, 1]$
- 2 **Allen–Cahn**: $\partial_t u = \varepsilon \Delta u + u - u^3$,
 $u \in [-1, 1]$
- 3 **Burgers**: $\partial_t u = \nu \partial_{xx} u - u \partial_x u$,
 $u \in [-1, 1]$

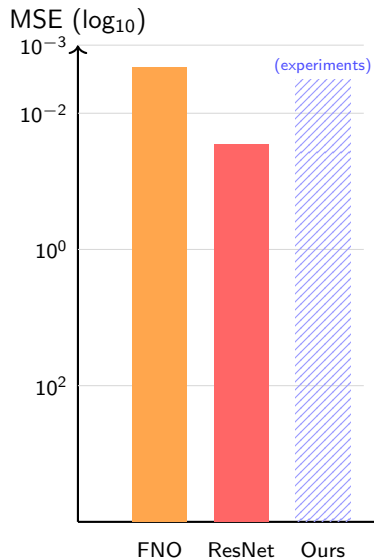
Configuration

- $N = 64$ spatial nodes, periodic BC
(2D extension: 32×32 , in progress)
- 1000 training / 50 validation trajectories
- RK4 with $M = 30$ substeps per τ
- AdamW, lr= 10^{-3} , cosine decay

Baselines

- **ResNet**: 4-block residual network
(41.6K params)
- **FNO**: 4-layer Fourier Neural Operator
(17.5K params)
- **Ours**: Latent semigroup (9.6K params)

Fisher-KPP: Short-Time Comparison ($T = 2$)



Key guarantees (by construction):

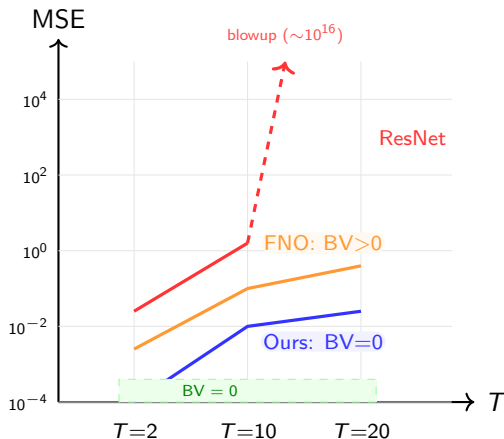
- ✓ Boundary violation: 0.00
- ✓ Energy monotone: 100%
- ✓ Semigroup defect: $\sim 10^{-14}$

Parameters: 9.6K

vs. baselines:

- FNO: BV 4.6×10^{-4} , defect 2.2×10^{-5}
- ResNet: BV 7.2×10^{-4} , defect 2.1×10^{-4}

Long-Time Rollout: Why Structure Matters



Key observation:

ResNet exhibits **numerical blowup** beyond $T = 10$.

FNO stays finite but boundary violations grow to 1.8%.

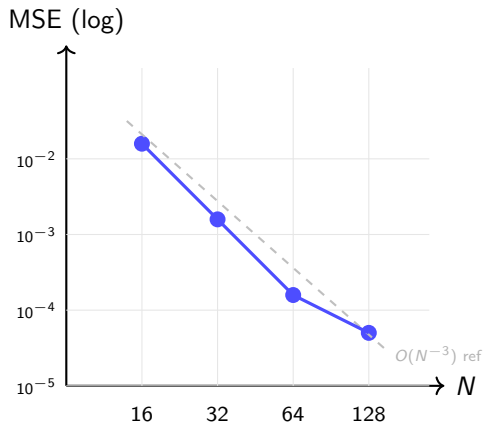
Ours maintains exact boundary compliance (0.00) at all times.

Why?

$\varepsilon_{\text{sg}} = 0$ (exact semigroup) prevents error accumulation.

Boundary compliance by construction (sigmoid map).

Spatial Convergence Study (Fisher–KPP)



Convergence data (experiments in progress):

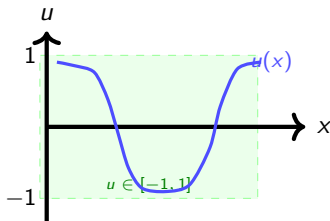
N	MSE
16	...
32	...
64	...
128	...

Expected behavior:

- Error decreases monotonically with N
- Effective ~ 3 rd-order for $N=16 \rightarrow 64$
- Rigorous rate analysis deferred

Allen–Cahn Benchmark

PDE: $\partial_t u = \varepsilon^2 \Delta u + u(1 - u^2), \quad u \in [-1, 1]$



$\varepsilon = 0.1, N = 64, \Delta t = 0.1, T_{\max} = 2.0$

Our guarantees (by construction):

✓ MSE: 2.21×10^{-2}

✓ BV: **0.00**

✓ Defect: $\sim 10^{-14}$

✓ Energy: **100%**

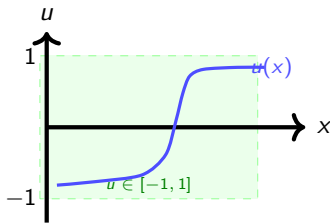
vs. baselines (experiments in progress):

- FNO/ResNet: ...

Architectural guarantees transfer to Allen–Cahn.

Viscous Burgers Benchmark

PDE: $\partial_t u + u \partial_x u = \nu \partial_{xx} u$, $u \in [-1, 1]$



$\nu = 0.01$, $N = 64$, $\Delta t = 0.05$, $T_{\max} = 1.0$

Our guarantees (by construction):

✓ BV: **0.00**

✓ Defect: $\sim 10^{-14}$

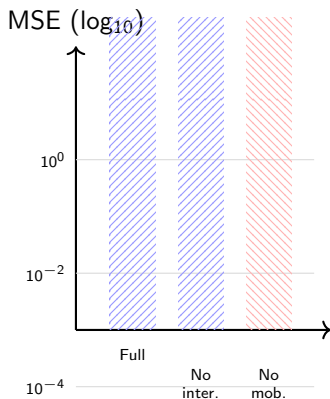
✓ Energy: **100%**

vs. baselines:

- FNO: BV 15.4%, defect 2.7×10^{-4}
- ResNet: BV 6.6%, defect 1.3×10^{-3}

Note: Burgers is *non-gradient* — expressivity guarantee does not formally apply, but architectural guarantees hold regardless.

Ablation Study (Fisher-KPP)



Findings (experiments in progress):

- Full & No-interaction: comparable MSE (reaction term dominates Fisher-KPP)
- **No-mobility**: significantly worse — $K_{\theta}(z)$ is the **critical component**

All variants: exact boundary compliance (0.00)
⇒ Architectural guarantee robust to component removal

Comparison with Classical Structure-Preserving Methods

Method	Known $\mathcal{F}$?	Guarantee
IRP schemes [Zhang & Shu, 2010]	Yes	Invariant region
SSP integrators [Gottlieb, Shu & Tadmor, 2001]	Yes	Convex constraints
Entropy-stable [Ju, Li & Qiao, 2022]	Yes	Energy dissipation
Ours	No (learned)	All three by construction

- Classical methods: design the *scheme* for a known PDE
- Ours: constrain the *learned vector field* itself
- Sigmoid map $\leftrightarrow$ positivity limiter; PSD mobility $\leftrightarrow$ dissipative integrator
- Key difference: we *learn* the generator within a structurally constrained model class

Summary and Take-Home Message

Core idea

Design	Guarantee
sigmoid map g	admissibility
$K \succ 0$ + gradient flow	dissipation
ODE flow	exact semigroup

Contributions

- 1 Semigroup-level error decomposition:
 $\|u^n - \mathcal{S}_{n\Delta t} u_0\| \leq \varepsilon_{\text{app}} + n \varepsilon_{\text{sg}}$
- 2 Architecture with $\varepsilon_{\text{sg}} = 0$ by construction
- 3 Transfer theorem + universal approximation

Open Problems

- 1 Global flow existence for $\beta_V = 0$
- 2 PDE-level transfer rigor
- 3 Expressivity for non-gradient generators
- 4 Extension to higher dimensions

**Physics should be built into
the architecture,
not just the loss function.**

Three structural laws $\rightarrow$ three architectural constraints

$\rightarrow$ guarantees hold for every θ , not just after

Thank You

Questions?

Preprint available upon request